

Jayesh Deepak Badgujar

Pune, Maharashtra, India | +91 82087 89115 | [linkedin.com/in/jayesh-badgujar](https://www.linkedin.com/in/jayesh-badgujar) | github.com/JayeshBadgujar18 | leetcode.com/u/deadlyjay

PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience across frontend engineering, backend systems, applied cryptography, and AI/ML infrastructure. Built a secure distributed payment-settlement backend using hybrid encryption and idempotent design, a vector database from scratch in C++ powering a full Retrieval-Augmented Generation (RAG) pipeline with a local LLM, and a responsive React application with modern state management. Seeking an internship or entry-level role to apply strong systems and full-stack engineering skills to real-world software.

TECHNICAL SKILLS

Languages: Java, C++, JavaScript, TypeScript, HTML5, CSS3

Frontend Development: React.js, Redux Toolkit, Vite, Tailwind CSS, Responsive UI Design

Backend & Systems: Spring Boot, RESTful API Design, H2/SQL Databases, Multithreading & Concurrency, Idempotent System Design, Data Structures & Algorithms (HNSW, KD-Tree)

AI / Machine Learning: Retrieval-Augmented Generation (RAG), Vector Databases, Ollama (Local LLMs), Text Embeddings, Similarity Search (Cosine, Euclidean, Manhattan), PCA Visualization

Security & Cryptography: RSA-OAEP, AES-256-GCM, Hybrid Encryption, Secure Distributed Messaging, Distributed Mesh Networking

Tools & Platforms: Git & GitHub, Vercel, REST API Testing, Maven, npm

PROJECTS

VectorDB – Vector Database Built From Scratch with RAG Pipeline — C++, HNSW, Ollama, REST API

2026

[GitHub Repo](#)

- Built a fully functional vector database from scratch in C++, implementing HNSW, KD-Tree, and Brute Force search algorithms with cosine, Euclidean, and Manhattan distance metrics.
- Integrated a local LLM pipeline (Ollama) for real-time text embedding generation and a full Retrieval-Augmented Generation (RAG) workflow to answer questions from user-provided documents.
- Developed a REST API and interactive web UI with live 2D PCA visualization of vector space and side-by-side algorithm/speed benchmarking.
- Engineered the system to run entirely offline and locally, replicating core mechanics behind production vector databases like Pinecone and Weaviate.

UPI Offline Mesh – Offline UPI Payment Simulation via Bluetooth Mesh Network — Java, Spring Boot, H2, Cryptography

2026

[GitHub Repo](#)

- Designed and built a backend simulating offline UPI payments routed through a Bluetooth-style mesh network, enabling encrypted transactions to reach a server without live internet connectivity.
- Implemented hybrid RSA-OAEP + AES-256-GCM encryption so payment data stays unreadable and tamper-proof while passing through untrusted intermediary devices.
- Solved distributed idempotency using atomic compare-and-set hashing, validated with concurrent tests simulating 3 bridge nodes delivering the same packet simultaneously — exactly one settles, duplicates are safely dropped.
- Built a software-based mesh simulator with gossip-protocol relay logic and a live dashboard to demonstrate the full encrypt-relay-settle pipeline end-to-end.

NoteSaver – Pastebin-Style Note Sharing Web App — React, Vite, Redux Toolkit, Tailwind CSS

2025

[GitHub Repo](#) | [Live Demo](#)

- Built a lightweight pastebin-style application allowing users to create, view, and manage text snippets ("pastes") through a clean, responsive UI.
- Implemented global state management using Redux Toolkit for predictable and scalable data handling across components.
- Styled the interface with Tailwind CSS for a fast, responsive, mobile-friendly design.
- Leveraged Vite for fast development builds and Hot Module Replacement (HMR); deployed the production build on Vercel.

EDUCATION

Bachelor of Engineering – Information Technology

Sinhgad College of Engineering, Pune, Maharashtra

CERTIFICATIONS & COURSES

- AWS Cloud Practitioner Essentials – Amazon Web Services (In Progress)
- The Complete Web Development Bootcamp – Udemy
- Docker & Kubernetes: The Practical Guide – Udemy

ACHIEVEMENTS & ACTIVITIES

- Active contributor on GitHub with consistent commit history across full-stack, systems, and AI/ML projects.
- Participated in college-level hackathons focused on cloud-native application development.
- Member of the college technical committee – organized workshops on Git, Docker, and Linux fundamentals.